Stabilizing differentiable partial optimal transport through sliced approaches

Sliced optimal transport scales by projecting distributions onto random lines, where transport is solved in closed form; partial optimal transport relaxes mass conservation to compare distributions of unequal mass. Combining the two is attractive but unstable: the optimally retained mass varies from one slice to the next, producing noisy gradients once aggregated. Taming this instability is the goal of this internship.

§1. Organization

- **Start date**: As soon as possible (flexible).
- **Localization**: IRISA, Rennes.
- **Supervision**:
  - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
  - Laetitia Chapel (TAGADA research group, IRISA, Vannes).

§2. Context

Optimal transport (OT) provides a principled geometry for comparing probability distributions, but its cubic cost limits its use at scale. *Sliced* OT sidesteps this by projecting the distributions onto many random one-dimensional lines, solving the resulting transport problems in closed form via sorting, and averaging. *Partial* OT, in turn, relaxes the mass-conservation constraint so that only a fraction of the mass needs to be transported — valuable when distributions differ in total mass or contain outliers (Bonneel & Coeurjolly, 2019; Bai et al., 2023).

Recent work makes both ingredients efficient: partial Wasserstein distances on the line admit a fast solution (Chapel & Tavenard, 2025), and sliced transport *plans* can be made differentiable through a bilevel reformulation (Chapel et al., 2025). Sliced partial and unbalanced variants have also been proposed (Bai et al., 2023; Séjourné et al., 2023). What remains poorly understood is the *differentiability* of the combined object.

§3. Scientific challenge

The difficulty is structural, not merely numerical. In a partial one-dimensional problem the solver must decide *which* mass to keep and which to discard; this retained set is the solution of a combinatorial selection that is *piecewise constant* in the projection direction. As a slice rotates, the active set switches at discrete angles, and at each switch the per-slice transport cost is continuous but its gradient jumps. In balanced sliced OT the optimal coupling varies smoothly and these jumps are absent, so Monte-Carlo averaging over slices yields low-variance gradients; in the partial case the discontinuities do *not* cancel under averaging, and the back-projected gradient is biased and high-variance, with the effect growing as the mass-imbalance ratio and the ambient dimension increase. The central question is therefore: **can the support-selection step be relaxed so that sliced partial OT becomes smoothly differentiable, without destroying the very mass-discarding behaviour that makes partial OT useful?** Quadratic or entropic regularization of the selection (Nguyen et al., 2025) is one promising route, but its interaction with slicing — and whether it preserves a valid metric — is open.

§4. Goals

Concretely, the candidate will:

1. reproduce and *quantify* the support-selection instability, relating per-slice gradient discontinuities to back-projected gradient variance as a function of mass-imbalance ratio, ambient dimension, and slice count;
2. design and compare stabilization strategies — soft/entropic or quadratic relaxation of the selection, slice reweighting, and consistent aggregation across projections — assessing which preserve metric/approximation guarantees;
3. benchmark against exact (non-sliced) partial OT on problems small enough to admit ground truth, measuring both value and gradient accuracy;
4. validate the stabilized estimator as a differentiable loss in a downstream training task (e.g. robust matching or generative alignment), reporting stability and convergence.

Candidates should possess:

- good Python coding skills,
- knowledge of the fundamentals of optimal transport and numerical optimization,
- familiarity with the POT (Python Optimal Transport) library is a plus,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a practical and theoretical understanding of sliced and partial optimal transport,
- the ability to diagnose and stabilize gradient estimators,
- experience benchmarking differentiable losses for downstream training.

§5. Bibliography

Bai, Y., Schmitzer, B., Thorpe, M., & Kolouri, S. (2023). *Sliced Optimal Partial Transport.* CVPR.

Bonneel, N., & Coeurjolly, D. (2019). *SPOT: Sliced Partial Optimal Transport.* ACM Transactions on Graphics.

Chapel, L., & Tavenard, R. (2025). *One for All and All for One: Efficient Computation of Partial Wasserstein Distances on the Line.* ICLR.

Chapel, L., Tavenard, R., & Vaiter, S. (2025). *Differentiable Generalized Sliced Wasserstein Plans.* NeurIPS. arXiv:2505.22049.

Nguyen, K., et al. (2025). *Sparse Partial Optimal Transport via Quadratic Regularization.* arXiv:2508.08476.

Séjourné, T., Bonet, C., Fatras, K., Nadjahi, K., & Courty, N. (2023). *Unbalanced Optimal Transport meets Sliced-Wasserstein.* arXiv:2306.07176.